

Stage Report II: Independent Prospective Cohort (C1) Lock, Execution and Verification Status

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Stage	Content	Status	Property	Value	Note
S1	Reproduction and controlled diagnostics	complete	Assets	20	pre-named universe, no addition
S2	Structural validation of the candidate regularity	FAIL	Provider	Yahoo	one provider only; no composite
S3	Capacity-controlled breadth panel (F1)	SUCCESS	Window	2016-01-04/01 – 2025-12-31	daily OHLC, unadjusted
S4	Independent prospective cohort (C1)	executing	Exclusions	0	all 20 delivered and accepted
TABLE I			Rows per asset	2444–2602	market calendar differences
PROGRAMME SEQUENCE. EACH STAGE IS GATED BY THE PREVIOUS ONE; S4 IS THE SUBJECT OF THIS REPORT.			Hash table	per file	verified on intake

Abstract—This report records the *execution status* of the independent prospective cohort study (C1) as of 2026-09-14: a locked twenty-asset, single-provider cohort; a predictor frozen before any outcome exists; a 120-run panel delegated to a peer machine that is offline-capable and idempotent; and the intake protocol that accepts or rejects each returned batch. It deliberately contains no analysis of C1 outcomes.

Index Terms—prospective validation, pre-registration, blinded analysis, cohort locking, execution provenance, intake verification.

I. PURPOSE AND SCOPE

The programme has moved from exploratory diagnostics to a *prospective* test. This report documents the state of that test’s *machinery* and nothing else.

II. PROGRAMME SEQUENCE AND CURRENT POSITION

Four stages have been executed in order, each gated by the previous one:

S1 established the reproduction and the operator-specialisation diagnostics. S2 tested the candidate structural regularity on four pre-registered assets and returned a failure, triggered by a capacity contradiction: an asset-level operator preference whose sign is not stable across model widths. S3 then asked the more basic question and answered it in the narrow sense that asset-level preference replicates on a majority of a seven-asset panel while remaining a *tendency*, not a property of an asset. S4 — the present stage — tests the candidate association *prospectively* on a genuinely independent cohort.

III. THE LOCKED COHORT

The cohort was locked before any outcome was inspected, and the lock is a table of hashes rather than a list of intentions.

Two properties matter for what follows. The cohort shares the *provider* of the discovery set while differing in *market*: its assets exclude both the original ten and

TABLE II
COHORT LOCK SUMMARY. THE PER-FILE SHA-256 TABLE IS THE AUTHORITATIVE RECORD; ROW COUNTS DIFFER ACROSS ASSETS BECAUSE EXCHANGE CALENDARS DIFFER, NOT BECAUSE DATA WERE IMPUTED.

the four assets used in S2. Rows were dropped only where the provider returned all-null price rows and were never interpolated, with the drop counts documented per file. Acquisition was performed by the peer machine, because the analysing host cannot reach the provider.

IV. THE FROZEN PREDICTOR

The predictor is frozen *before* any outcome exists, which is the difference between a prospective test and a descriptive one. Its construction rule:

- a feature window covering closes $[i, i + 96)$ is usable only if the whole window ends at or before the decision point, i.e. $i + 96 \leq \text{border1_test}$;
- the decision point follows the frozen split convention, $\text{border1_test} = n - \lfloor 0.2n \rfloor - 96$ with $\text{num_train} = \lfloor 0.7n \rfloor$;
- the asset-level predictor aggregates the window features over the usable windows only, so the last close index touched is $\text{border1_test} - 1$.

A leakage check is executed and printed: it asserts that every index used by the predictor lies strictly below the decision point. The check passed for all twenty assets. A contemporaneous analogue computed over the test windows is carried as a *contrast only* and is never used for the primary test. Because the predictor is frozen, it may not be recomputed with a different window, estimator or cutoff once outcomes are seen.

V. DELEGATED EXECUTION

S4 requires 120 training runs (20 assets $\times$ 2 arms $\times$ 3 seeds) under the unchanged protocol, with per-run predictions and ground truth retained for the paired analysis.

Property	Value	Note	#	Milestone	Status
Runs	120	$20 \times 2 \times 3$	M1	Cohort acquired by the peer machine	done
Concurrency	2 lanes	frozen; widening declined	M2	Intake validation of the cohort	done, 20/20
Threads per process	14	unchanged mid-campaign	M3	Cohort locked with a per-file hash table	done
Retained per run	metrics, pred, true	command line and log tail	M4	Predictor frozen with a leakage check	done, all assets
Checkpoints	not transferred	not needed for the analysis	M5	Verifier hardened and smoke-tested	done
Batch reporting	40 runs or 6 h	manifest per batch	M6	Panel launched (120 runs, 2 lanes)	running, 0/120
Progress at writing	0/120	launched 13:36, first runs	M7	First batch returned and verified	pending
TABLE III			M8	Blind third-party recomputation	pending

EXECUTION CONFIGURATION. THROUGHPUT CHANGES THAT WOULD ALTER RUN COMPARABILITY MID-CAMPAIGN (THREAD COUNT, LOADER WORKERS, EXTRA LANES) WERE DECLINED AND DEFERRED.

TABLE IV
MILESTONES. M7 GATES M8, AND NO STAGE BEYOND M7 IS ATTEMPTED BEFORE THE PANEL COMPLETES.

The executor was selected because its network reaches the provider while the analysing host does not. Its runner is *offline-capable* by construction: everything is staged once, after which training reads only local files, so a network outage cannot interrupt the campaign. Resilience rests on four mechanisms: idempotent skipping of completed runs; a watchdog that restarts lanes without ever duplicating a run; detection of stalled runs by *process-tree CPU time* rather than log output; and a contingency plan that pre-authorises self-recovery while forbidding any protocol or data change. One genuine defect was found and fixed during the campaign: a restart had orphaned two finished runs that no lane recorded, which would have produced a silent two-run gap; the runner now *ingests* such runs and marks them in the manifest, so the process, not merely the content, is auditable.

VI. INTAKE VERIFICATION

Nothing enters the evidence chain unverified. Each returned batch is checked mechanically before any file is accepted:

- 1) **Protocol conformance per run:** the stored command line must contain every frozen hyperparameter value together with that run’s own seed, model and data path.
- 2) **Content integrity:** predictions and ground truth must match the batch manifest hashes.
- 3) **Metric integrity:** the metric array must parse as the frozen six-value order.
- 4) **Whole-batch rejection:** if any run, manifest row, hash or command line disagrees, the *entire* batch is rejected and returned with a difference list; partial results are never copied. The verifier’s default mode is a dry run.
- 5) **Handoff inventory:** an accepted batch yields a per-file SHA-256 inventory and a handoff receipt recording the frozen revision actually executed.

One gap cannot be closed by the receiving side alone: transferred files can be shown to match the batch manifest, but only the executor can show that its local originals match what was transferred. That confirmation is therefore a required handoff item, not an optional courtesy.

VII. INDEPENDENCE AND GOVERNANCE

Three roles are kept separate: the *executor* runs the panel and returns raw artefacts without performing any analysis; the *analyst* computes the predictor and the headline statistic; a *blind third party*, which has seen neither the analysis nor its own expectations, recomputes the headline statistic from the raw artefacts and reports a difference list, with the explicit rule that a discrepancy is a finding and never something to be tuned away. Executor and analyst are both excluded from that third role, and readers of this report are asked not to treat it as evidence about C1’s outcome.

VIII. MILESTONES AND STATUS

IX. NEXT STEPS

- 1) Continue execution and health monitoring; keep the campaign’s conditions constant.
- 2) Verify each returned batch under the intake protocol; reject whole batches on any mismatch.
- 3) On completion, assemble the handoff: runs, prediction and ground-truth arrays, the batch manifest, the frozen revision actually executed, the per-file hash inventory, and the executor’s local-origin hash confirmation.
- 4) Release the artefact set to the blind third party and await its difference list; only then read the pre-committed interpretation of whichever branch the outcome falls into.

X. WHAT THIS REPORT DELIBERATELY OMITTS

No estimate of the headline statistic appears here, no per-asset interaction effect, no reading of either branch of the pre-committed interpretation, and no claim about the eligibility of any structured operator. Those omissions are not gaps: they are the condition under which the report is useful, and the same applies to any progress summary written while the panel runs.

REPRODUCIBILITY NOTES

All artefacts cited here live in the project repository: the lock table with per-file hashes, the frozen predictor, the intake verifier with its inventory and handoff receipt, and the batch manifests that reproduce each run’s command line.